

The 2026 Capstone Project

AURA

3D Brain Tumor Segmentation

Attention · U-Net · Residual · All-modality

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Advisor: Dr. Basanta



Outline

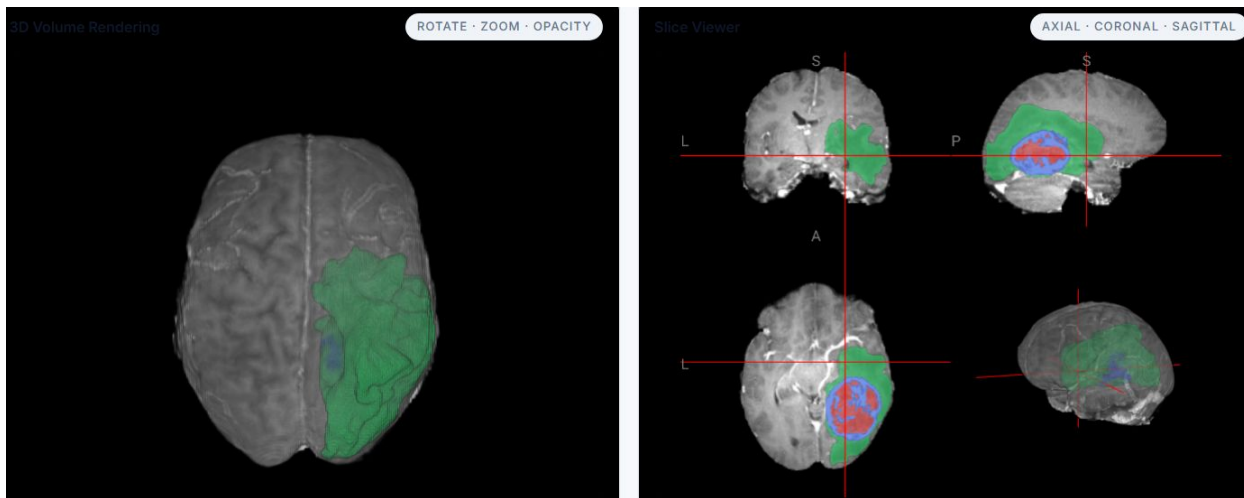
- Motivation
- Dataset
- Methodology
- Results
- Deployment
- Future Work
- Conclusion
- Q & A

01

Motivation

The Problem — and AURA

- Brain tumors are highly fatal and 5-year survival is ~ 30%
- Manual MRI segmentation takes up to 4 hours
- Lack of **3D visualization** and instant **volume metrics**



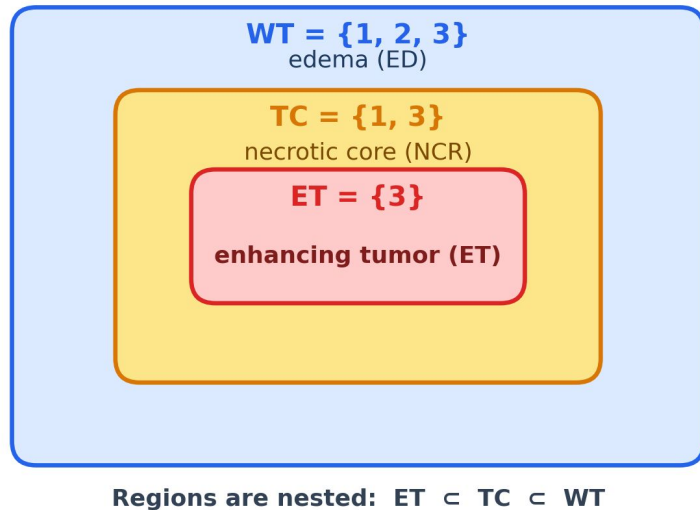
AURA Workstation

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Dataset

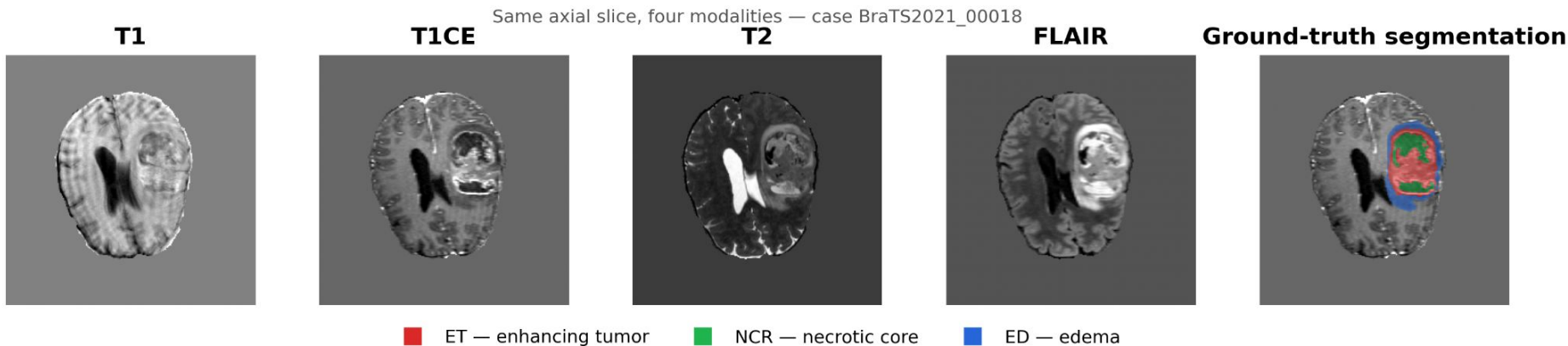
BraTS Overview

- Dataset: BraTS 2021
- Modalities: T1, T1CE, T2, FLAIR
- Classes: 0 BG, 1 NCR, 2 ED, 3 ET
- Train / Val: 1000 / 251 cases
- Shape: $240 \times 240 \times 155$
- Input: 5-ch, 128^3 patches



MRI Modality

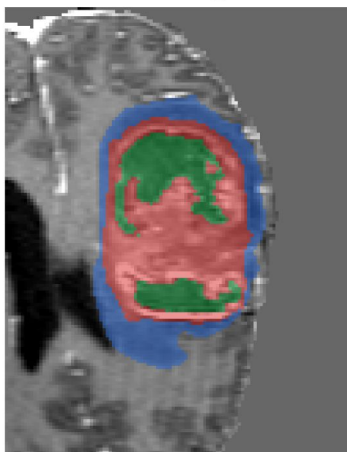
- No single modality shows the whole tumor
- **T1CE** highlights enhancing tumor
- **FLAIR** highlights edema
- **T1 / T2** give anatomical structure



Clinical Region

- $\text{NCR} = \text{dead core}$ • $\text{ET} = \text{active tumor}$ • $\text{ED} = \text{infiltrative spread}$
- Resect ET and NCR to remove the tumor mass and relieve pressure; radiate ED for eliminate infiltrative cells

Real MRI scan — the three regions inside one tumor



■ NCR — necrotic core ■ ET — enhancing rim ■ ED — infiltrative edema

Tumor Core (TC) → Surgery

$$\text{TC} = \text{NCR} + \text{ET} \quad (\text{solid tumor mass})$$

Resect to debulk the tumor and relieve pressure

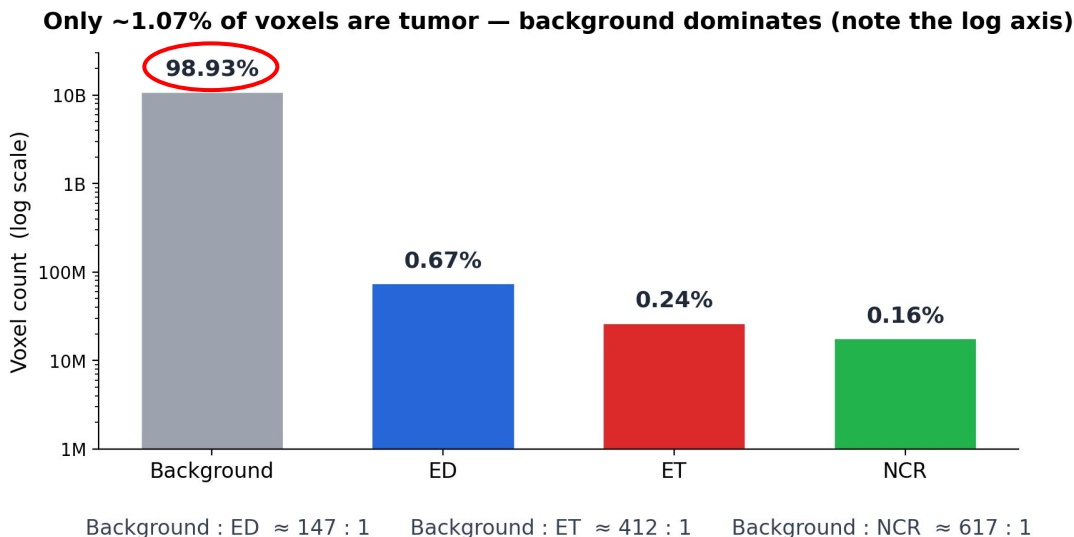
Whole Tumor (WT) → Radiotherapy

$$\text{WT} = \text{NCR} + \text{ET} + \text{ED} \quad (\text{mass} + \text{infiltration})$$

Irradiate the field to kill infiltrating cells

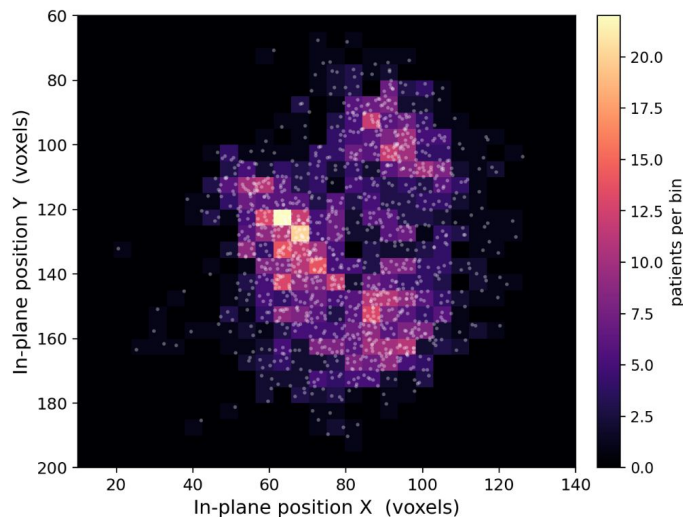
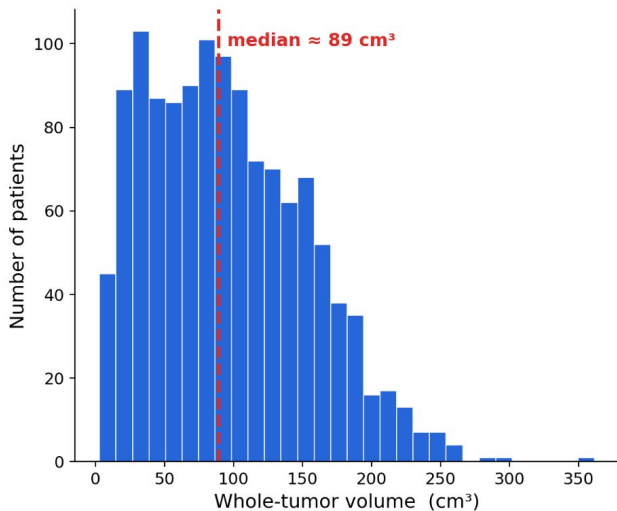
Class Imbalance

- Background : tumor $\approx 99 : 1$; ET alone $\approx 412 : 1$
- CE collapses to background, so we use region-wise Dice + Focal
- Same reason 50% of training patches are tumor-centered



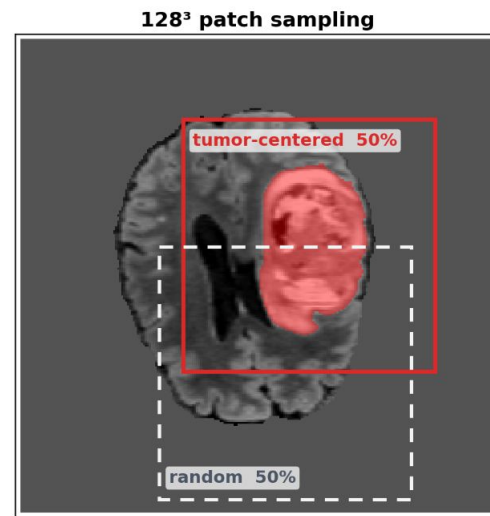
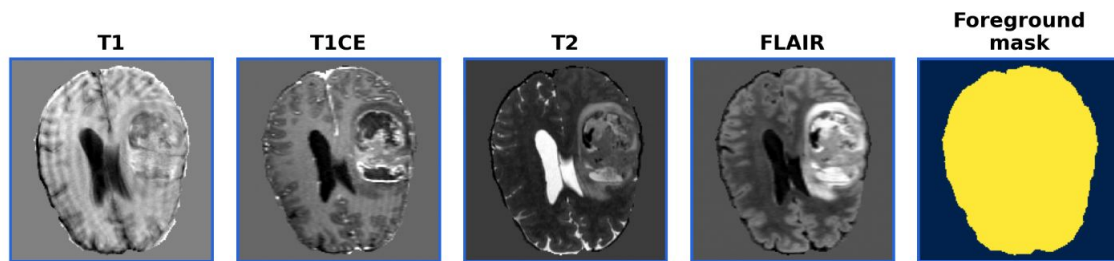
Tumor Sizes & Location

- Volume spans $\sim 3\text{--}362\text{ cm}^3$ — a $\sim 129\times$ range (median $\approx 89\text{ cm}^3$)
- Centroids scattered brain-wide — **no positional shortcut**
- Leads to patch-based training + sliding-window inference



Model Input

- Input is **5-channel**: T1 · T1CE · T2 · FLAIR + a **foreground** mask
- Z-score on brain voxels per modality, background zeroed
- Train on 128^3 patches — 50% tumor-centered, 50% random

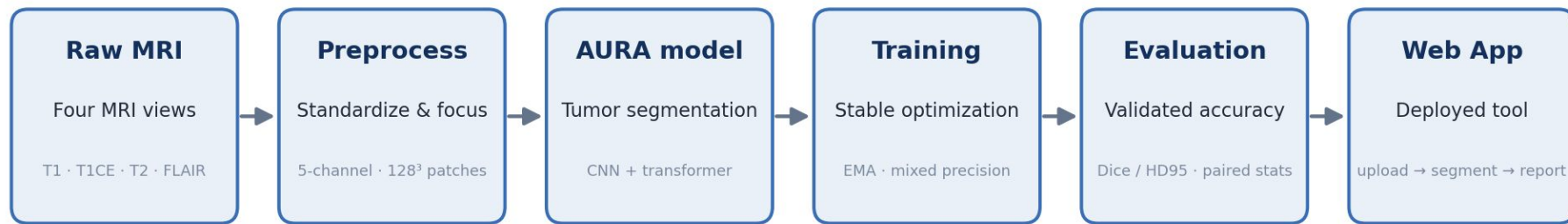


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Methodology

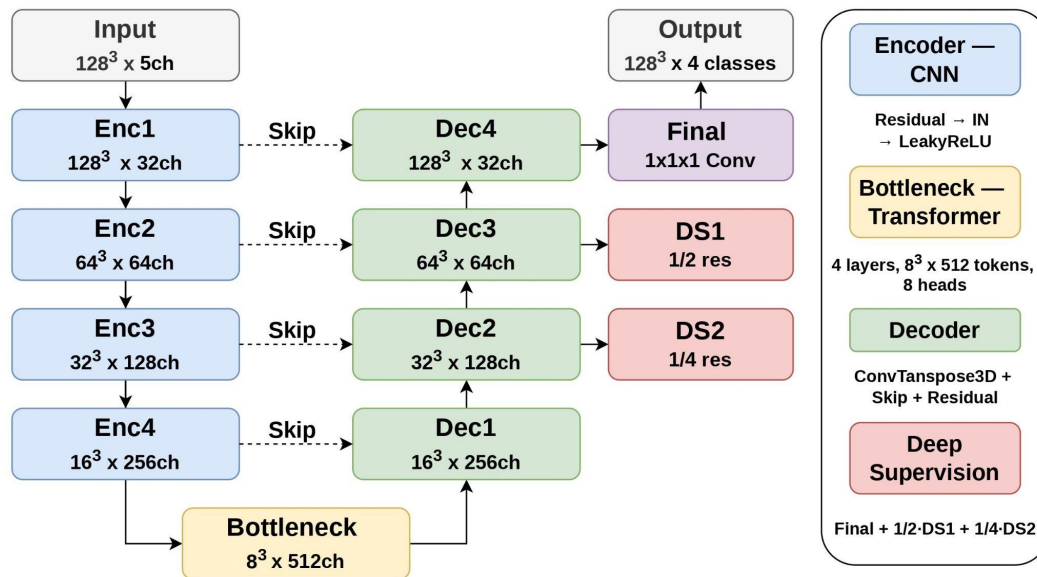
System Pipeline Overview

AURA is built as a complete end-to-end **medical AI system**, not just a model. Every stage from raw MRI to deployed web workstation is built, trained, and evaluated together.



AURA Architecture

- **CNN + Transformer** hybrid captures local texture and global 3D context
- Transformer at the 8^3 **bottleneck only**, efficient and fp16-stable
- **4-class softmax** avoids region-overlap ambiguity of sigmoid outputs



Loss Function

- **Region-wise Dice + Focal** loss over ET, TC, WT
- Focal loss focuses training on hard, misclassified voxels
- Deep supervision propagating gradients through decoder stages

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{seg}}(\hat{y})}_{\text{full res}} + \underbrace{\frac{1}{2} \mathcal{L}_{\text{seg}}(\hat{y}_{DS_1})}_{1/2 \text{ res}} + \underbrace{\frac{1}{4} \mathcal{L}_{\text{seg}}(\hat{y}_{DS_2})}_{1/4 \text{ res}}$$

$$\mathcal{L}_{\text{seg}} = \sum_{r \in \{\text{WT}, \text{TC}, \text{ET}\}} (\text{Dice}_r + \text{Focal}_r)$$

$$\text{Dice}_r = 1 - \frac{2 \sum p_r g_r}{\sum p_r + \sum g_r} \quad \text{Focal}_r = - \sum (1 - p_t)^\gamma \log p_t, \quad \gamma = 2$$

Training Recipe

Stable optimization for 3D medical image segmentation.

Component	Setting
Optimizer	Adam, lr 1e-4
Effective batch size	32 (batch 2 × accum 16)
Epochs	300
AMP	fp16
EMA	decay 0.999
Gradient clipping	max norm 1.0
Patch sampling	50% tumor-centered

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Results

Comparison Models

Baseline

Modules:

3D Residual U-Net

Role:

Lower bound

Complex

Modules:

3D Residual U-Net

+ Modality

+ Cross modal attention

+ Spectral Swin

+ Frequency filter

+ Uncertainty head

+ Boundary head

+ Multi-scale fusion

Role:

Upper bound

AURA

Modules:

3D Residual U-Net

+ Transformer bottleneck

Role:

Hero model

Evaluation Protocol

- AURA is evaluated across accuracy, calibration, and uncertainty
- TTA – 8-way flip, axes (D, H, W)
- MC-Dropout – 20 forward passes with Dropout active

Category	Metric	Measures
Segmentation	Dice	Overlap between prediction and ground truth
Segmentation	HD95	Worst-case boundary error (mm)
Calibration	ECE	How well confidence matches accuracy on tumor voxels
Uncertainty	AURC	Whether the model knows when it is wrong

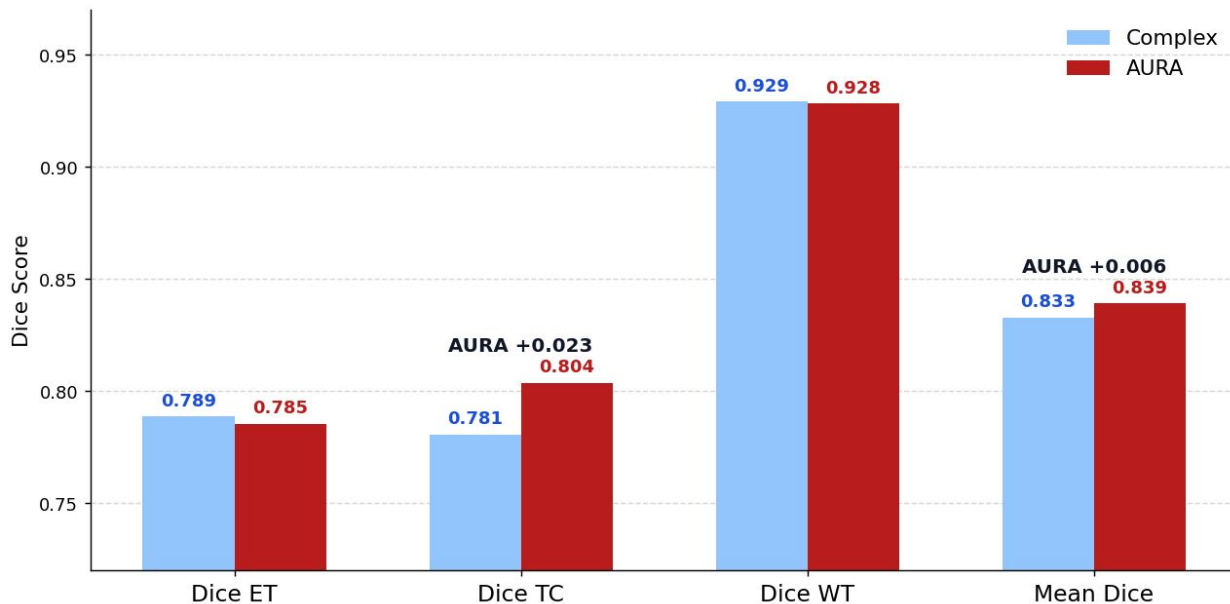
Evaluation Result — AURA Wins

- AURA achieves the **highest Mean Dice** (0.839), outperforming both Baseline and Complex
- Best calibration and uncertainty — **lowest ECE** (0.100) and **AURC** (0.175) across all models

Model	Mean Dice	Dice ET	Dice TC	Dice WT	HD95 ↓	ECE ↓	AURC ↓
Baseline	0.825	0.765	0.788	0.923	6.750	0.106	0.179
Complex	0.833	0.789	0.781	0.929	5.95	0.110	0.191
AURA	0.839	0.785	0.804	0.928	5.96	0.100	0.175

Less is More

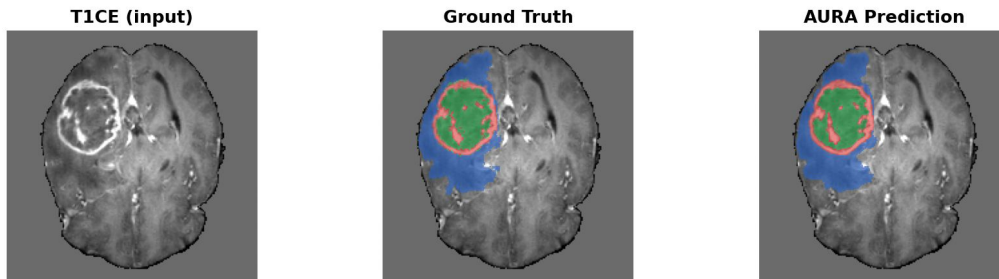
- **More modules $\neq$ better performance**, Complex has 8, AURA has 2
- AURA wins TC by +0.023 – the most clinically critical region



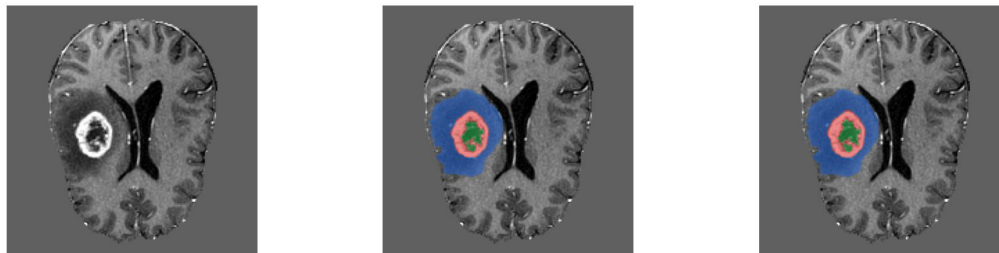
Qualitative — Success

AURA correctly identifies all three tumor regions across cases.

Patient BraTS2021_01418 | Mean Dice 0.92 · Tumor 187,569 voxels



Patient BraTS2021_01593 | Mean Dice 0.99 · Tumor 77,802 voxels

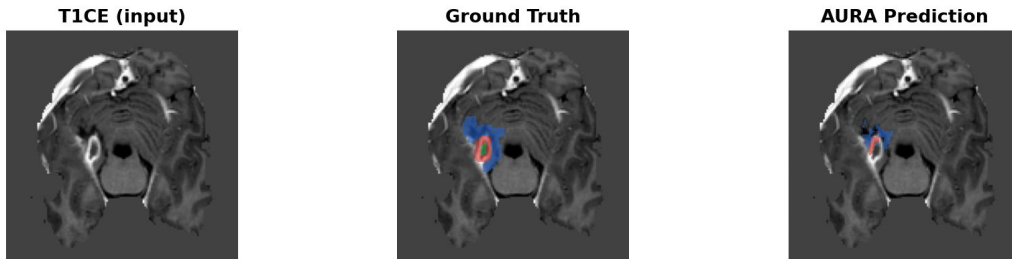


■ NCR ■ ED ■ ET

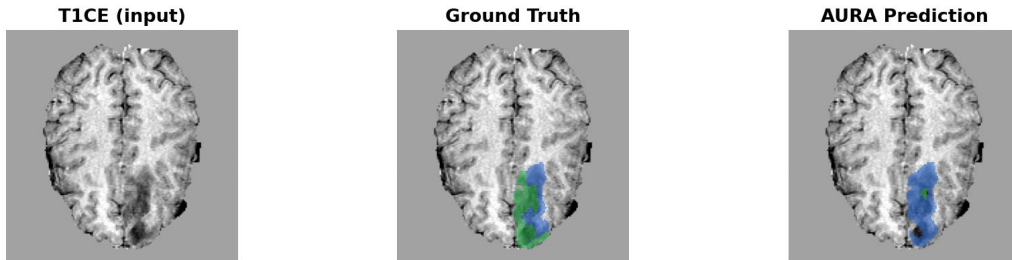
Qualitative — Failure

AURA struggles when ET is small and NCR is misread as edema.

Patient BraTS2021_01628 | ET Dice ≈ 0 · TC Dice 0.25 · Tumor 2,808 voxels



Patient BraTS2021_01508 | ET Dice ≈ 0 · TC Dice 0.02 · Tumor 35,851 voxels

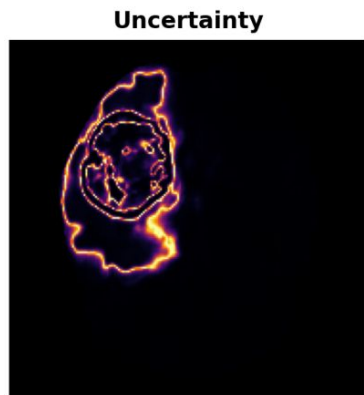


■ NCR ■ ED ■ ET

Trustworthiness

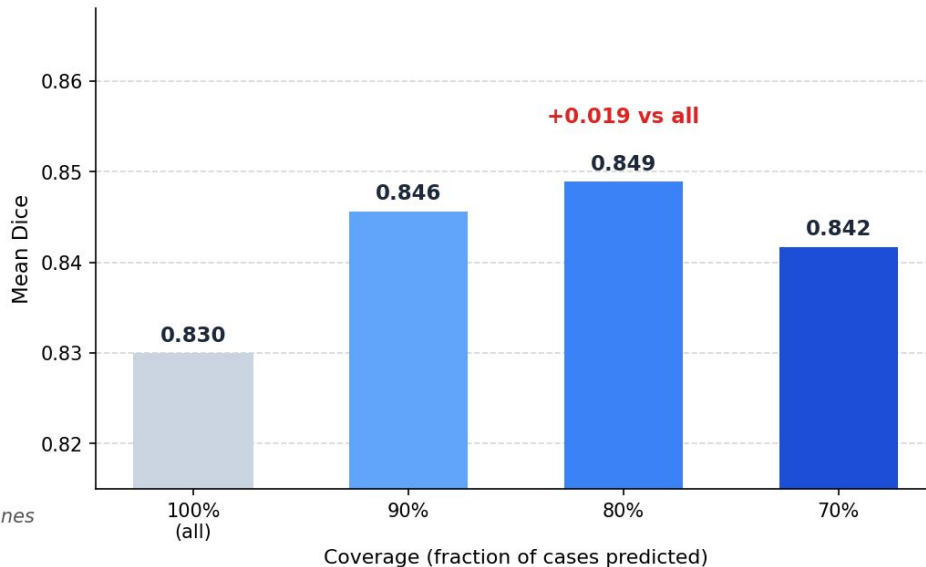
- AURA assigns lower confidence to cases it is likely to get wrong
- Skipping the least confident 20% raises Dice from 0.830 to 0.849

Uncertainty Map (BraTS2021_01418)



High uncertainty concentrates at region boundaries and error zones

Selective Prediction (TTA)



Complexity

At **45M** parameters and **242 ms** per scan, AURA fits on a single consumer GPU with no compromise on accuracy.

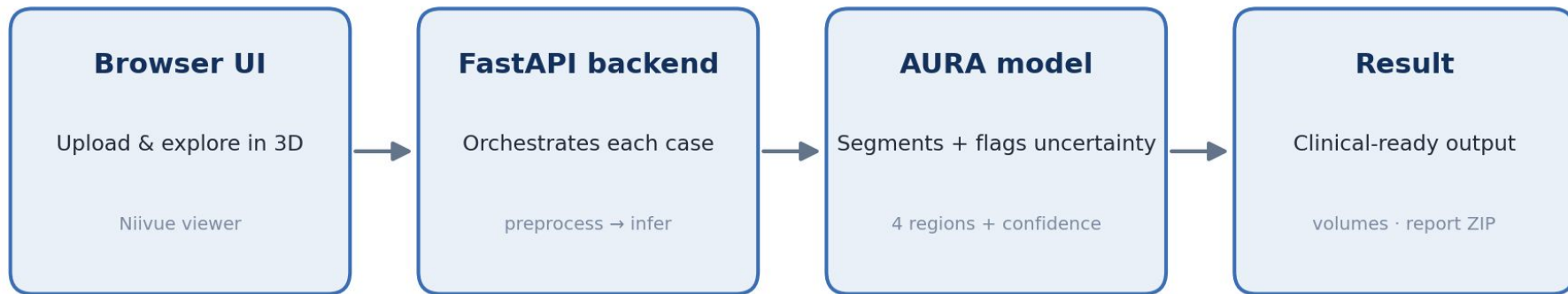
Metric	AURA
Parameters	45 M
FLOPs	612 GFLOPs
GPU Memory	1.9 GB
Model Size	172 MB
Inference Time	242 ms
Throughput	4.1 cases/s

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Deployment

The Web Workstation

- **A real tool, not just a model** — AURA running live in a browser
- Capabilities: **3D view · volumes · confidence · anatomy · report**



Live Demo



Brain Tumor Segmentation

Clinical-style MRI workstation · Attention · U-Net · Residual · All-modality

Model: AURA

1 Upload MRI

Four NIFTI modalities (.nii / .nii.gz)

T1 ✓ BraTS2021_01418_t1.nii

T1CE ✓ BraTS2021_01418_t1ce.nii

T2 ✓ BraTS2021_01418_t2.nii

FLAIR ✓ BraTS2021_01418_flair.nii

▶ Run Prediction

Done.

2 View Controls

Base modality

T1CE (contrast)

Overlay opacity



- Necrotic Core (NCR) — red
- Edema / swelling (ED) — green
- Enhancing Tumor (ET) — blue

NCR is the inner dead-tissue core — it sits *inside* ET/ED, so it is mostly hidden in the 3D render but visible in the slice viewer.

3 Export



Clinical Summary

One-paragraph readout — copy directly into a medical record.

Predicted whole tumor volume 193.8 mL (95th percentile, Very High), tumor core 59.5 mL, enhancing tumor 43.3 mL. Tumor primarily involves Insula_L (9%), Frontal_Sup_2_L (9%), Frontal_Mid_2_L (8%). Mean per-region confidence: WT 0.98, TC 0.99, ET 0.96.



Tumor Volumes

Total tissue affected, in millilitres (mL)

Whole Tumor
All abnormal tissue (NCR + ED + ET) **193.8 mL**

Tumor Core
Necrotic + enhancing tissue **59.5 mL**

Enhancing Tumor
Active, contrast-enhancing region **43.3 mL**



Model Confidence

Mean predicted probability inside each region (0–100%)

Whole Tumor
Very high certainty **98%**

Tumor Core
Very high certainty **99%**

Enhancing Tumor
Very high certainty **96%**



Size vs Population

Whole-tumor volume compared with 1251 BraTS patients

VERY HIGH

94.6th percentile · larger than 94.6% of the cohort

Below the 33rd percentile → Low · up to 67th → Medium · up to 90th → High · above → Very High.



Anatomical Location

Where the tumor sits in the brain — top 5 regions it overlaps (AAL3 atlas), with what each region normally does

Left Insular cortex **INSULA** **9.2%**
interception, taste & emotion
Insula_L

Left Putamen **BASAL GANGLIA** **5.1%**
movement regulation & motor skills
Putamen_L

Left Superior frontal gyrus **FRONTAL LOBE** **8.6%**
working memory & planning
Frontal_Sup_2_L

Left Inferior frontal gyrus **FRONTAL LOBE** **4.8%**
speech production (Broca's area, left)
Frontal_Inf_Tri_L

Left Middle frontal gyrus **FRONTAL LOBE** **8%**
attention & executive function
Frontal_Mid_2_L

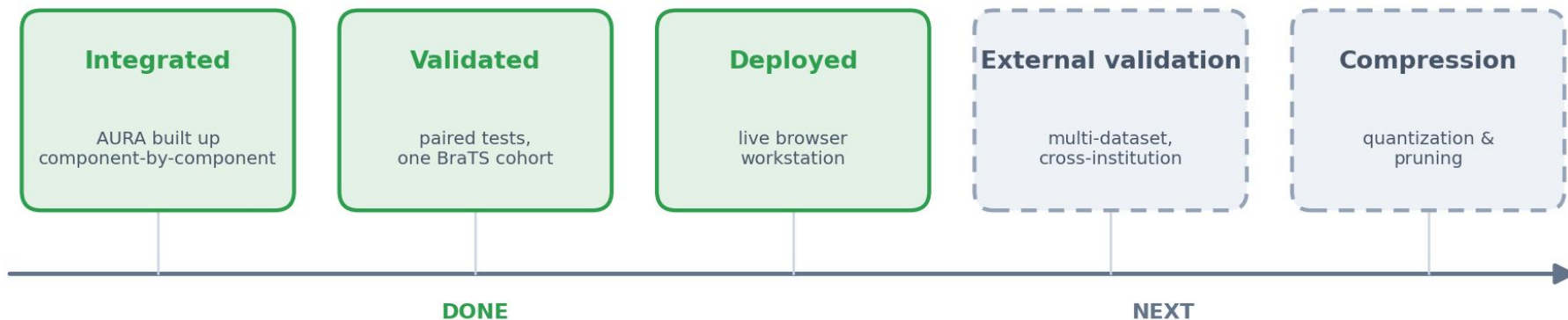
Each percentage is the share of the whole-tumor mask that falls inside that labelled region. **Left / Right** is the brain hemisphere; the small grey code (e.g. *Calcarine_R*) is the raw atlas label. Localisation is approximate — the AAL atlas is in MNI152 space while BraTS scans are in SRI24 space.

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Future Work

Future Work

- Validate AURA across other datasets and institutions
- **Quantize** and **prune** the model for faster inference



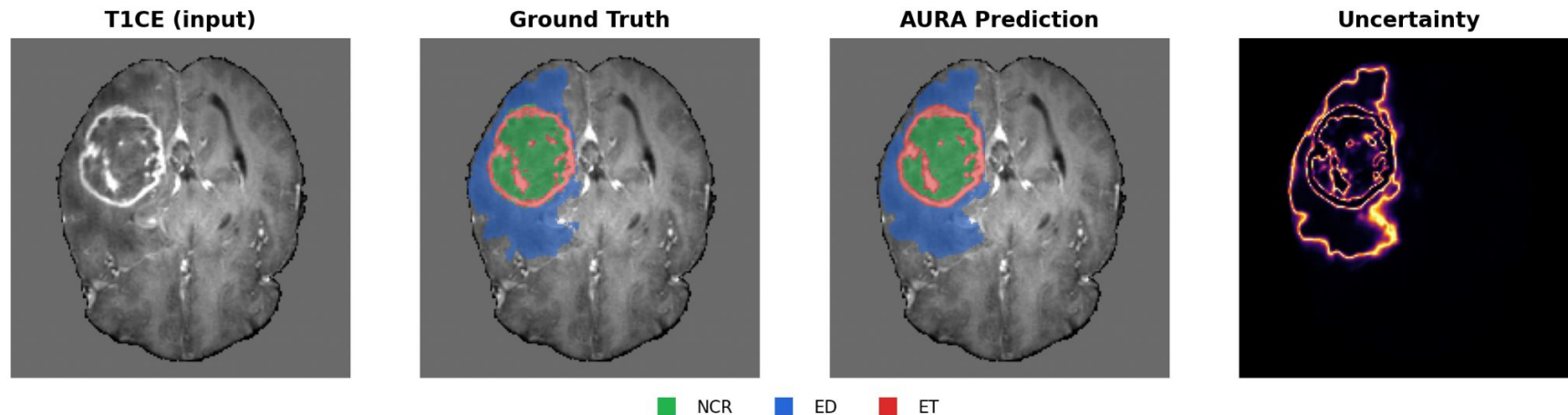
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Conclusion

Conclusion

- **Accurate** — strong segmentation across ET, TC, WT
- **Trustworthy** — calibrated uncertainty that tracks real error
- **Deployed** — the validated model, live in the browser

Representative case BraTS2021_01418 — AURA prediction matches the expert, and uncertainty flags the errors



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Q & A

Thank You for Your Time

Live Demo Video

📁

Brain Tumor Segmentation

Clinical-style MRI workstation · Attention · U-Net · Residual · All-modality

Model: AURA

1 Upload MRI

Four NIFTI modalities (.nii / .nii.gz)

T1

Choose a .nii / ...

Browse

T1CE

Choose a .nii / ...

Browse

T2

Choose a .nii / ...

Browse

FLAIR

Choose a .nii / ...

Browse

▶ Run Prediction

2 View Controls

Base modality

T1CE (contrast)

Overlay opacity

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NCR is the inner dead-tissue core — it sits *inside* ET/ED, so it is mostly hidden in the 3D render but visible in the slice

Clinical Summary

One-paragraph readout — copy directly into a medical record.

Upload a 4-modality MRI study to generate a report.

Tumor Volumes

Total tissue affected, in millilitres (mL)

Whole Tumor

All abnormal tissue (NCR + ED + ET)

Tumor Core

Necrotic + enhancing tissue

Enhancing Tumor

Active, contrast-enhancing region

Model Confidence

Mean predicted probability inside each region (0–100%)

Whole Tumor

Tumor Core

Enhancing Tumor

Size vs Population

Whole-tumor volume compared with 1251 BraTS patients

Awaiting prediction

Below the 33rd percentile → Low · up to 67th → Medium · up to 90th → High · above → Very High.

Anatomical Location

Where the tumor sits in the brain — top 5 regions it overlaps (AAL3 atlas), with what each region normally does

Upload a study to compute regional involvement.

Each percentage is the share of the whole-tumor mask that falls inside that labelled region. **Left / Right** is the brain hemisphere; the small grey code (e.g. *Calcarine_R*) is the raw atlas label. Localisation is approximate — the AAL atlas is in MNI152 space while BraTS scans are in SRI24 space.

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Feng-Yu Hsu